

Nmap Network Scan Report

Task 02 — Network Scanning & Service Enumeration

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Scan Date	27 June 2026
Target System	localhost (127.0.0.1) — Personal Kali Linux VM
Nmap Version	7.98
Report Type	Authorized Internal Scan — Educational Purpose

DISCLAIMER: This scan was performed solely on the tester's own Kali Linux virtual machine for educational and internship purposes. No unauthorized systems were scanned. All activities comply with ethical hacking principles.

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1. Objective

The primary objective of this task is to gain practical hands-on experience with the second phase of the ethical hacking methodology — Scanning and Enumeration. Using Nmap (Network Mapper), this exercise demonstrates how security professionals identify active hosts, detect open ports, enumerate running services, and gather operating system information on a target system in a safe, controlled, and fully authorized environment.

All scans in this report were performed exclusively against the loopback address (127.0.0.1 / localhost) of the tester's own Kali Linux virtual machine. No external or third-party systems were targeted at any point.

2. Scope and Environment

Parameter	Details
Target	localhost / 127.0.0.1
Platform	Kali Linux (x86_64-pc-linux-gnu) — Oracle VirtualBox VM
Nmap Version	7.98 (https://nmap.org)
Compiled Libraries	liblua-5.4.8, openssl-3.5.5, libssh2-1.11.1, libz-1.3.1, libpcr2-10.46, libpcap-1.10.6, nmap-libdnet-1.18.0, ipv6
Available Engines	epoll, poll, select
Scan Authorization	Self-authorized — personal virtual machine only
Scan Date & Time	27 June 2026, between 18:32 and 18:38 IST (+0530)

3. Commands Executed

The following Nmap commands were executed as observed in the provided screenshots:

Command	Purpose
<code>nmap --version</code>	Verified that Nmap was correctly installed and displayed version details.
<code>nmap localhost</code>	Performed a default TCP SYN scan on all 1000 common ports of the loopback address.
<code>nmap -sV localhost</code>	Ran service version detection to identify what software was running on open ports.
<code>sudo nmap -0 localhost</code>	Attempted OS detection — failed with an 'unrecognized option -0' error (zero instead of capital letter O).

<code>sudo nmap -o localhost</code>	Second OS attempt — also produced no results; Nmap reported 0 hosts scanned (lowercase -o is an output flag, not OS detection).
<code>sudo nmap -A localhost</code>	Ran aggressive scan combining OS detection, service version detection, script scanning, and traceroute.

4. Scan Results

4.1 — Default Scan: `nmap localhost`

Executed at 18:32 IST. Scanned the loopback interface (127.0.0.1).

Field	Value
Host Status	Up (0.0000020s latency)
IPv6 Address (not scanned)	::1
Ports Scanned	1000 (default TCP common ports)
Open Ports Found	0 (None)
Closed Ports	1000 (all reset)
Scan Duration	0.08 seconds

4.2 — Service Version Scan: `nmap -sV localhost`

Executed at 18:32 IST. Attempted to detect service versions on open ports.

Field	Value
Host Status	Up (0.0000020s latency)
Open Ports Found	0 (None)
Closed Ports	1000 (all reset)
Services Detected	None
Scan Duration	0.32 seconds
Note	Service detection performed — no services found to enumerate

4.3 — Aggressive Scan: `sudo nmap -A localhost`

Executed at 18:38 IST. The -A flag enables OS detection, version detection, script scanning, and traceroute.

Field	Value
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Host Status	Up (0.000036s latency)
Open Ports Found	0 (None)
Closed Ports	1000 (all reset)
OS Detection Result	Too many fingerprints match — no specific OS identified
Network Distance	0 hops
Scan Duration	1.87 seconds

5. Open Ports and Services

Across all scans performed — including the default scan, service version scan, and aggressive scan — no open TCP ports were detected on localhost (127.0.0.1). All 1000 scanned ports returned a 'closed' (reset) state.

Port Range Scanned	Services Found	Status
Top 1000 TCP Ports (default Nmap list)	None	All Closed (Reset)

This result indicates that no network services were actively listening on the loopback interface at the time of scanning. On a freshly installed or minimal Kali Linux VM with no services manually started (e.g., SSH, Apache, MySQL), this is the expected and correct behaviour.

6. Operating System Detection

6.1 — Failed Attempts (Incorrect Flags)

Two OS detection attempts were made with incorrect flags:

`sudo nmap -0 localhost` — Nmap returned an error: 'unrecognized option -0'. This happened because the digit zero (0) was used instead of the capital letter O.

`sudo nmap -o localhost` — This ran but produced no scan results because the lowercase -o flag is used for specifying an output file, not for OS detection. Nmap reported 'WARNING: No targets were specified, so 0 hosts were scanned.'

6.2 — Aggressive Scan OS Detection Result (`sudo nmap -A localhost`)

The -A flag includes OS detection (equivalent to -O). Nmap was able to reach the host but returned the following message:

```
"Too many fingerprints match this host to give specific OS details"
```

This means Nmap collected OS fingerprint data from the loopback interface, but the fingerprint matched too many OS profiles to determine a single definitive match. This is a common occurrence when scanning the loopback address, as the local stack behaves differently from a real network interface. The network distance was confirmed as 0 hops, confirming it is the local machine.

Why is OS Detection Important in Penetration Testing?

- It helps the tester understand the target environment and select appropriate exploit payloads.
- Different operating systems have different vulnerability profiles and patch schedules.
- Knowing the OS version narrows down which CVEs (Common Vulnerabilities and Exposures) may apply.
- It assists in planning privilege escalation paths during post-exploitation.
- OS fingerprinting can reveal mismatches (e.g., a server claiming to be one OS but behaving like another), which may indicate honeypots or deception technologies.

7. Errors and Observations

Flag Typo (-0 vs -O): The command 'sudo nmap -0 localhost' failed because the digit zero was used instead of the capital letter O for OS detection. The correct flag is 'sudo nmap -O localhost'.

Incorrect Flag (-o): 'sudo nmap -o localhost' was misinterpreted as an output specification, resulting in no scan being performed. Nmap reported 0 hosts scanned.

No Open Ports: All 1000 TCP ports on localhost were found to be in a closed (reset) state. This is expected on a minimal Kali VM where no services have been started.

IPv6 Address Not Scanned: Nmap reported ::1 as an additional address for localhost but did not scan it by default. A separate scan with 'nmap -6 ::1' would be needed to enumerate IPv6.

OS Fingerprint Ambiguity: The aggressive scan detected TCP/IP stack characteristics but could not narrow down the OS to a single match. This is typical behavior when scanning via the loopback interface.

Scan Speed: The default scan completed in 0.08 seconds and the aggressive scan in 1.87 seconds — demonstrating the additional overhead of OS and service detection.

8. Security Recommendations

#	Recommendation	Details
1	Use Correct Nmap Flags	Always double-check Nmap flags before running scans. The OS detection flag is a capital letter O (-O), not the digit zero (-0). Refer to 'nmap -h' or the official documentation.
2	Run Services Only When Needed	Even though no open ports were found, it is good practice to enable services (SSH, HTTP, etc.) only when actively using them and disable them afterward.
3	Scan IPv6 Separately	Nmap does not automatically scan IPv6 addresses. Use 'nmap -6 ::1' to check the IPv6 loopback interface as well.
4	Use -O for Dedicated OS Detection	For more reliable OS detection on a real target, use 'sudo nmap -O ' on a network interface rather than the loopback address.

5	Perform Regular Self-Audits	Regularly scan your own systems to identify any unintended open ports or services that may have been started automatically after software installations.
6	Combine Scans for Complete Picture	Use 'sudo nmap -A -p- ' to scan all 65535 ports with aggressive detection for a thorough audit, rather than relying solely on the default top-1000 port scan.

9. Conclusion

This task provided a foundational, hands-on understanding of the scanning and enumeration phase of ethical hacking. Using Nmap version 7.98 on a Kali Linux virtual machine, a series of increasingly detailed scans were performed against the loopback interface (127.0.0.1). The scans confirmed that the host was active but had no open TCP ports across the top 1000 default Nmap port list — an expected result on a minimal VM with no services running.

An important learning from this session was the significance of using precise command-line flags. The attempt to run OS detection with '-0' (zero) instead of '-O' (capital letter O) produced an error, highlighting how small syntax mistakes can cause scans to fail entirely. Similarly, running '-o' (lowercase) triggered an unrelated output-file behaviour rather than OS fingerprinting.

The aggressive scan (-A) demonstrated that even without open ports, Nmap attempts to gather OS fingerprint information. Though it could not conclusively identify the operating system due to loopback interface limitations, it confirmed a network distance of 0 hops — proving the target was the local machine itself.

Overall, this exercise reinforced the importance of the scanning phase in any security assessment. Understanding what is — and what is not — running on a system is critical before any deeper evaluation. The absence of open ports in this scan is itself a meaningful result, confirming the system's minimal attack surface in its current state.